

Highlights

Parallel lines have the same slope.

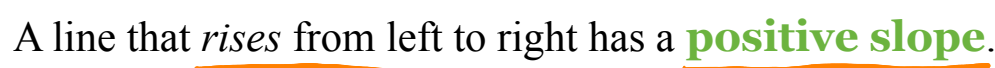
On the coordinate grid below, which two lines are parallel?



$$\text{Line } EF: m = \frac{-2 - (-4)}{3 - 2} = \frac{2}{1} = 2$$

Line $\overline{AB}$ and line $\overline{EF}$ have the same slope. They are parallel.

Look at the perpendicular lines graphed on the coordinate grid. Can you see that they go in opposite directions?



A line that *falls* from left to right has a **negative slope**.

Find the slopes of these two perpendicular lines.

Because the lines go in opposite directions, one slope is positive and the other slope is negative. The fractions are also inverted. This type of opposite is called a **negative reciprocal**. $-\frac{2}{3}$ is the negative reciprocal of $\frac{3}{2}$.

For perpendicular lines, the slope of one line is the negative reciprocal of the slope of the other line.

 Report Content Errors

Rate Your Confidence

^ High

Medium

Low

Category

Lesson 4: Problem Solving on the Coordinate Plane

 **Related Questions:**
[Practice questions taken related to this topic](#)